Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 23)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows the mass of carbon in the main reservoirs of the carbon cycle.

Table 1

Reservoir	Mass of carbon / billion tonnes
Atmosphere (CO_2)	700
Living organisms	470
Dead organic matter (soil)	3 700
Surface ocean waters	500
Deep ocean (dissolved)	35 000
Fossil fuels	10 000
Sedimentary carbonate rocks	20 000 000

0 1 . 1 Express the mass of carbon in the atmosphere in standard form, in tonnes.

[1 mark]

Answer ____ tonnes

0 1 . 2 Calculate the percentage increase in atmospheric carbon that would result from burning 10% of the fossil fuel reservoir, assuming all the carbon remained in the atmosphere.

[2 marks]

Answer ____ %

0 1 . 3 Explain why the sedimentary carbonate rocks, although by far the largest reservoir, have little effect on atmospheric CO_2 concentrations.

[2 marks]

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0 1 . 4 Describe how carbon moves from the atmosphere into sedimentary carbonate rocks.

[3 marks]

0 1 . 5 Explain why aerobic respiration returns carbon to the atmosphere as CO_2 but anaerobic respiration returns it mainly as methane.

[2 marks]

10

0 2 . 1 A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate ☐

B The readings are accurate but not precise ☐

C The readings are both accurate and precise ☐

D The readings are reliable because the meter is electronic ☐

0 2 . 2 Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

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0 2 . 3 Explain why a preliminary study is carried out before the main investigation.

[2 marks]

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0 3 . 1 Explain **three** features of fossil fuels that made them ideal for the development of industrial society.

[3 marks]

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0 3 . 2 75 t of aviation fuel carries a Boeing 747 and its 400 passengers from London to New York. Calculate the mass of fuel used per passenger, in kg.

[2 marks]

Answer _____ kg

0 3 . 3 Explain why the total quantity of fossil fuels in existence is much greater than the quantity counted as **reserves**.

[3 marks]

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0 3 . 4 Cheap North Sea gas contributed to the closure of the UK's deep coal mines, although large coal deposits remained. Explain why those mines cannot now be reopened.

[2 marks]

0 3 . 5 Explain **four** environmental impacts of the **extraction** of fossil fuels, as distinct from burning them.

[5 marks]

15

0 4 Table 1 shows four general approaches to pollution control.

0 4 . 1 Complete Table 1 by giving one example of each approach.

[4 marks]

Table 1

Approach	Example
Production prevention	
Prevention of release	
Post-release remediation	
Alternative processes	

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0 4 . 2 Explain why prevention of production is generally preferable to post-release remediation.

[2 marks]

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0 4 . 3 Describe how each of the following new technologies helps to control pollution.

[4 marks]

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Satellite monitoring of oil spills .....

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GPS ship tracking .....

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Polymer adsorption of heavy metals .....

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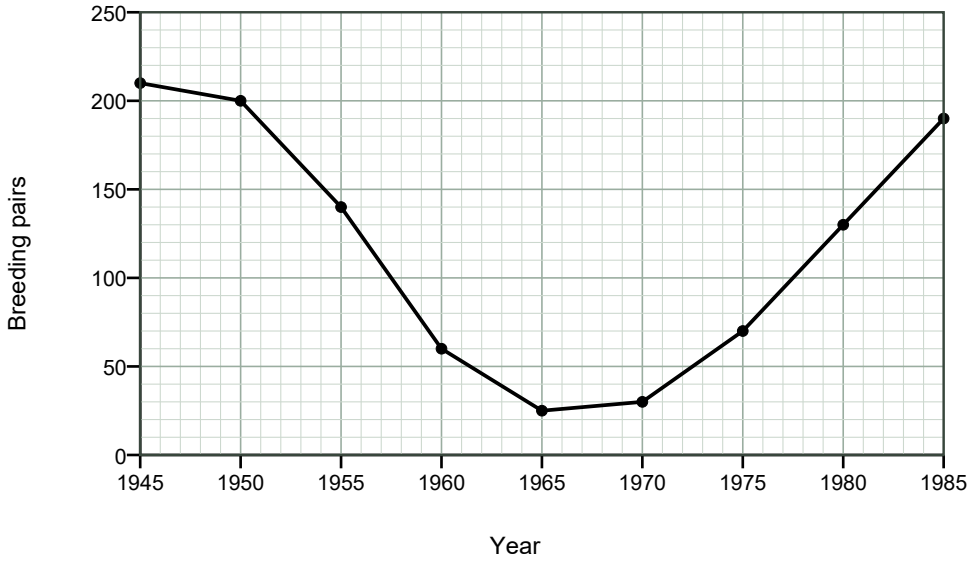
Phytoremediation .....

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10

0 5 Figure 1 shows the number of breeding pairs of sparrowhawks in a study area in the UK between 1945 and 1985, and the years in which DDT use began and was banned.

Figure 1



0 5 . 1 Describe the trend shown in Figure 1.

[2 marks]

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0 5 . 2 Calculate the percentage decrease in breeding pairs between 1950 and 1965. Show your working.

[2 marks]

Answer ____ %

0 5 . 3 DDT use in the UK began in the late 1940s and was restricted from the mid 1960s. Explain how the properties of DDT could account for the pattern in Figure 1.

[4 marks]

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0 5 . 4 Suggest why the recovery after 1970 was slow.

[2 marks]

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0 6 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 6 . 1 Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

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0 6 . 2 Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

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0 6 . 3 Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

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0 7 . 1 Describe how ozone is formed and destroyed naturally in the stratosphere.

[3 marks]

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0 7 . 2 Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).

[6 marks]

Useful properties

Properties causing ozone depletion

0 7 . 3 Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects.

[3 marks]

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0 7 . 4 State what is meant by an **ozone hole**.

[1 mark]

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0 7 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion.

[1 mark]

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0 7 . 6 Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

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15

0 8 . 1 Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

A High sand content ☐

B High clay content ☐

C Crumb peds ☐

D High pH ☐

0 8 . 2 Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

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0 8 . 3 Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

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0 9 . 1 Explain why it is difficult to be certain that a particular flood or heatwave is caused by anthropogenic climate change.

[3 marks]

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0 9 . 2 Explain why historic temperature records collected in towns may overestimate global warming.

[2 marks]

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0 9 . 3 Describe how each of the following sources of proxy data is used to estimate past climates and give one limitation.

[5 marks]

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Tree rings (dendrochronology) .....

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Ice cores .....

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1 0 . 1 Evaluate the risks and benefits of the uses of ionising radiation, and explain the principles used to control exposure of workers and the public.

[9 marks]

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1 0 . 2 Give **one** reason why the risks and benefits of nuclear electricity may not fall on the same people.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

OR

1 1 . 2 Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS